Lab-07: Map Network Exposure and Segmentation

Course: Cyber Security Awareness Workshop

Course code: C506

Time: 35 minutes

Alignment: LO1, LO3

Tool: [IP Calculator](https://alfredang.github.io/ipcalculator/)

Scenario

A retail office needs separate zones for staff, guests, point-of-sale devices and management systems.

Safety and authorisation boundary

- Use only the named training tool and the mock files in this folder.
- Do not enter real passwords, personal data, payment details or confidential information.
- Do not test any other website, network, account or organisation.
- Stop immediately if the target, data or behaviour differs from the written scope.
- Prefer reversible actions and preserve synthetic evidence.

Mock data

- `mock-network-assets.csv`
- `mock-network-requirements.csv`

Detailed procedure

1. Group the mock assets by trust level, data sensitivity and operational role.

2. Use the IP Calculator to create non-overlapping training subnets sized for each group.

3. Record network address, usable host range, broadcast address and proposed gateway for each zone.

4. Draw allowed flows and deny-by-default boundaries between staff, guest, point-of-sale and management networks.

5. Identify which services, if any, need internet exposure and add a monitoring or access-control requirement.

6. Review the design for excessive trust and document one residual risk.

Completion standard

- Subnet plan
- Asset-to-zone mapping
- Allowed-flow matrix
- Exposure decisions
- Residual risk

Reflection

What evidence would make you reverse your decision, and who would need to know?